

Earthquakes from 1965-2016: Data Analysis and Machine Learning Evaluation

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Introduction

Earthquakes, or quakes, occur when there is a sudden release of energy in the lithosphere that causes shaking on the Earth's surface. These seismic events occur in varying degrees of intensity and destruction, with an estimated 500,000 occurrences per year.^[1] The vast majority of these earthquakes are undetectable to humans without specialized equipment. Other earthquakes are far more intense and cause massive amounts of destruction to both life and property. Earthquake predication is an active area of study due to the direct impact on millions of people worldwide.

The focus of this project uses a database of earthquake occurrences from the National Earthquake Information Center^[2] from 1965-2016. The recorded earthquakes a minimum magnitude of 5.5. The database was acquired through Kaggle^[3]. Throughout this report we will explore the data, visualize data trends, and ultimately use machine learning principles to attempt to predict the magnitude category of earthquakes.

Methods

The methods used for analysis of the Earthquakes Database and machine learning (ML) evaluation follow the following pipeline:

1. Load data
2. Exploratory Data Analysis (EDA)
3. Data Cleaning
4. ML train/test data split
5. ML model training
6. Results evaluation
7. Iteration

Note: EDA and data cleaning steps were exchanged in this process, as there were a large number of columns with significant amounts of non-null values that were dropped prior to EDA.

Data loading and cleaning

Python and relevant libraries were the exclusive framework used for all steps of the ML pipeline throughout this project. The original, uncleaned dataset contained **21** feature columns and **23412** rows. The following table shows initial value information, including rows null-value count of the uncleaned dataset.

Column	Non-Null Count	Data Type
Date	23412 non-null	string

Column	Non-Null Count	Data Type
Time	23412 non-null	string
Latitude	23412 non-null	float64
Longitude	23412 non-null	float64
Type	23412 non-null	string
Depth	23412 non-null	float64
Depth Error	4461 non-null	float64
Depth Seismic Stations	7097 non-null	float64
Magnitude	23412 non-null	float64
Magnitude Type	23409 non-null	string
Magnitude Error	327 non-null	float64
Magnitude Seismic Stations	2564 non-null	float64
Azimuthal Gap	7299 non-null	float64
Horizontal Distance	1604 non-null	float64
Horizontal Error	1156 non-null	float64
Root Mean Square	17352 non-null	float64
ID	23412 non-null	string
Source	23412 non-null	string
Location Source	23412 non-null	string
Magnitude Source	23412 non-null	string
Status	23412 non-null	string

Note in the table above, there are a large number of columns that have a significant number of non-null values, meaning those values are entirely missing from the dataset, either because they were not recorded, or were omitted. Because this project is a relatively rudimentary ML exploration, columns with large amounts of non-null values, and other columns deemed irrelevant to this exploration were dropped, resulting in the following dataset:

Column	Non-Null Count	Data type
Date	23409 non-null	datetime64[ns]
Time	23409 non-null	datetime64[ns]
Latitude	23409 non-null	float64
Longitude	23409 non-null	float64
Type	23409 non-null	string

Column	Non-Null Count	Data type
Depth	23409 non-null	float64
Magnitude	23409 non-null	float64
Magnitude Type	23409 non-null	string
Source	23409 non-null	string
Location Source	23409 non-null	string
Magnitude Source	23410 non-null	string
Status	23409 non-null	string

During data cleaning the data type for Date and Time were also changed to datetime64 data type for ease of accessing data members during EDA.

Exploratory Data Analysis (EDA)

We begin the EDA with a simple histogram of all numeric columns in the dataset to visualize the distribution of those columns.

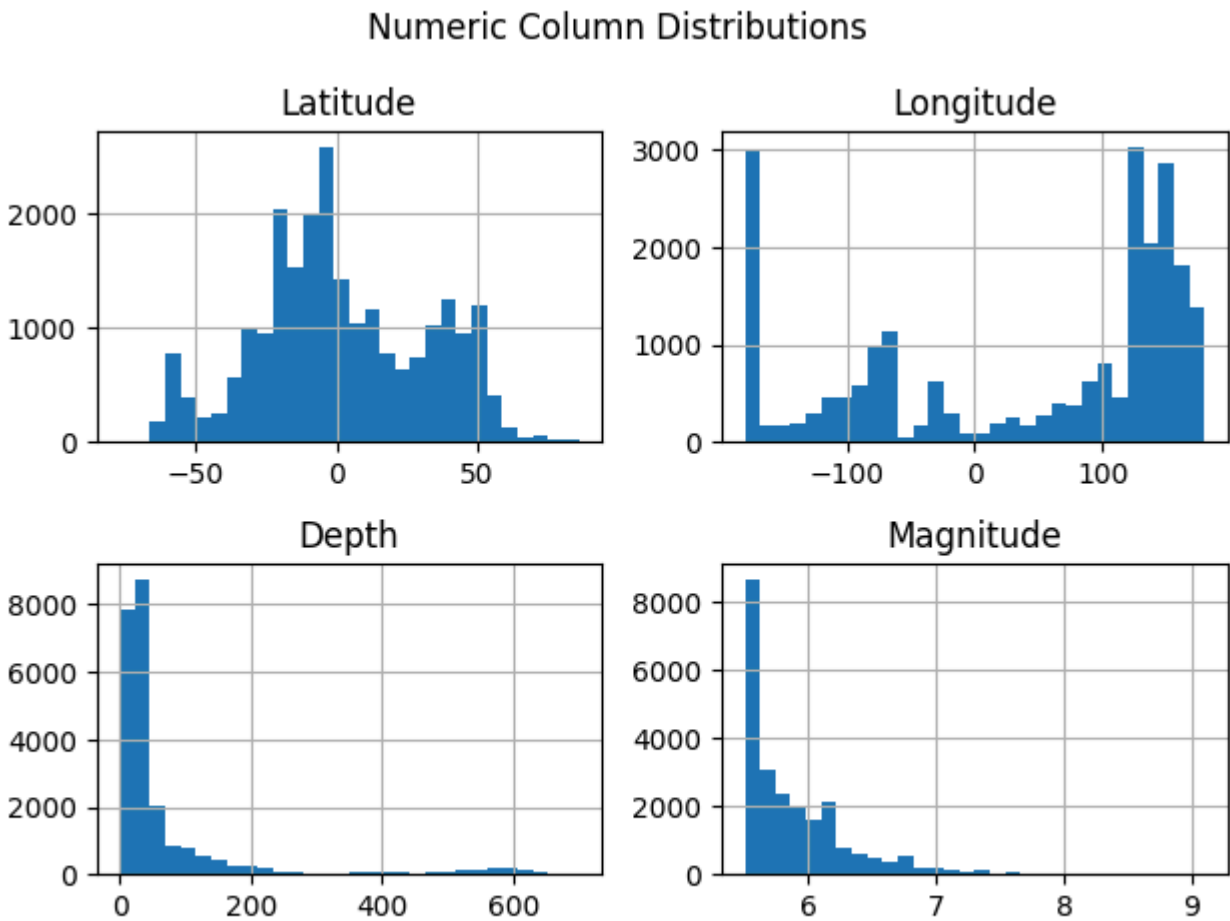


Figure 1: This figure gives a high-level view of distributions in the numeric columns of the earthquake dataset. Note the Depth and Magnitude columns are both skewed heavily to the right.

In addition to determining column distribution, it is useful to plot the earthquakes in a way that allows us to see geographic locations where earthquakes cluster. In the figure below, a scatter plot of latitudes and

longitudes for all earthquakes in the dataset was generate on top of a world map.

Global Earthquake Locations

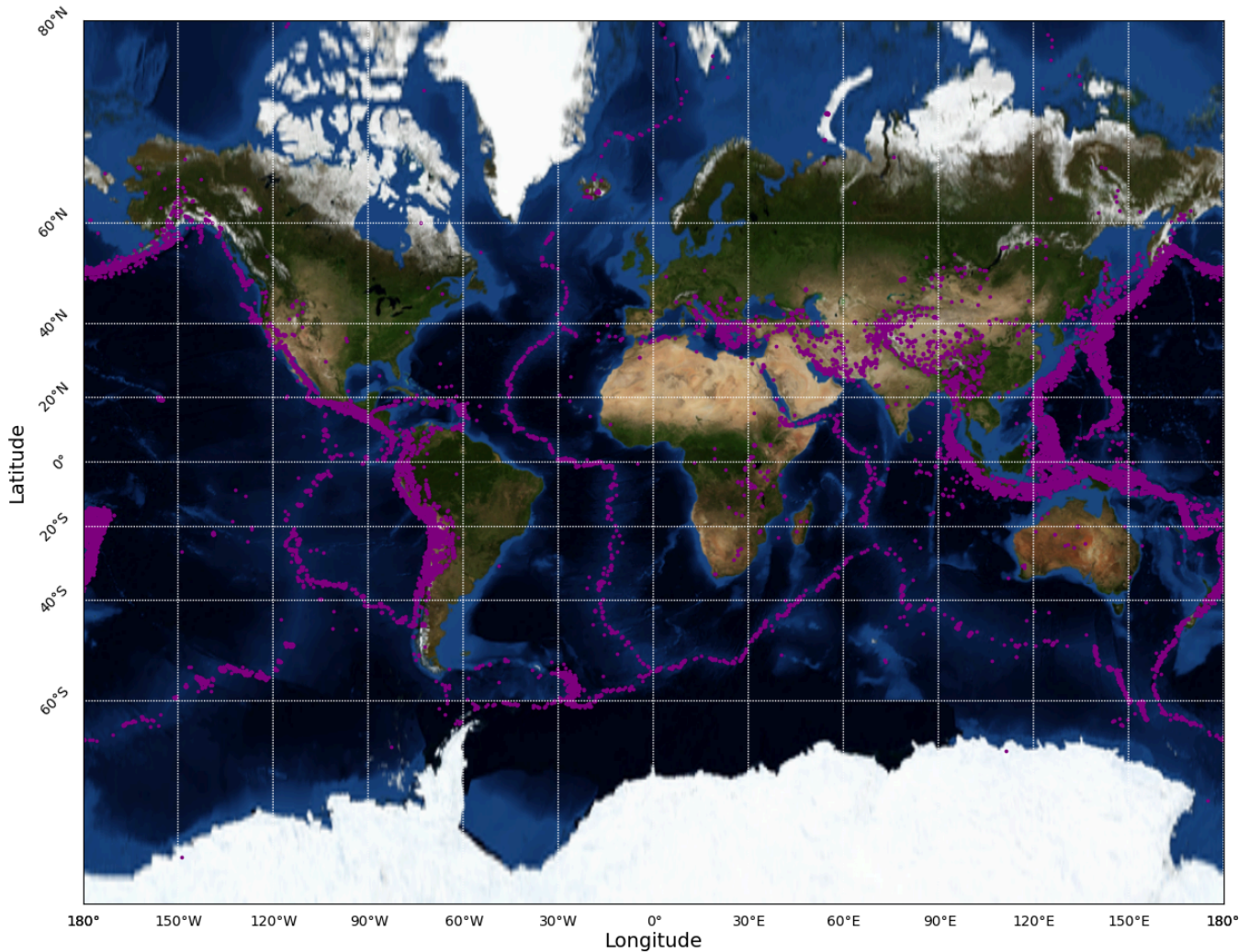


Figure 2: this figure is a scatter plot of all earthquakes from 1965-2016 with their latitude and longitude coordinates.

When comparing our scatter plot to the map of tectonic plates^[4] below, it's easy to see that the vast majority of earthquakes occur along the intersections of the tectonic plates around the world.



In the plot below we explore the relationship of earthquake depth to magnitude. While most of the less intense earthquakes are distributed quite equally throughout various depths, the strongest earthquakes (magnitude 8.0 and greater) clearly occur at much shallower depths.

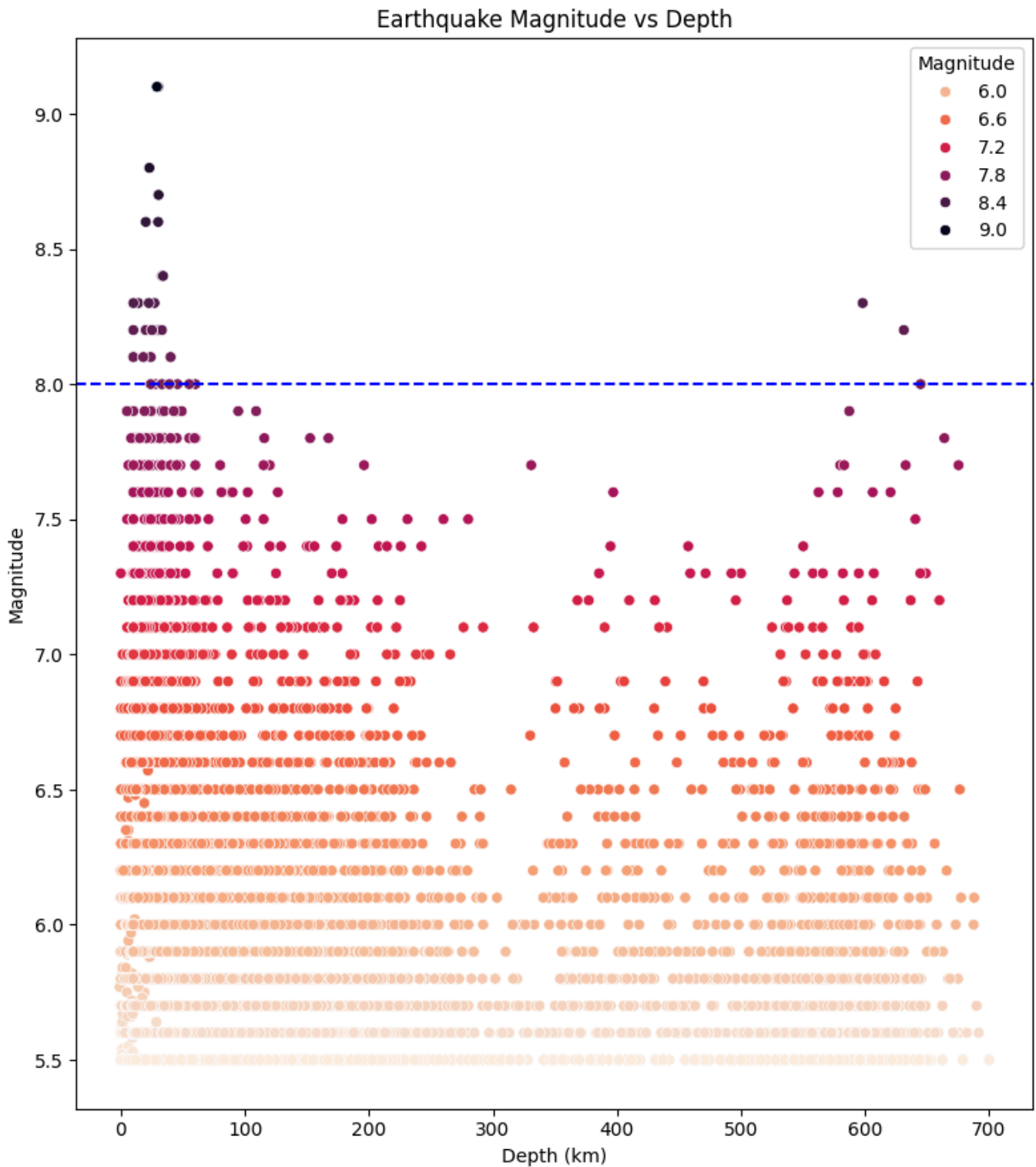


Figure 3: This scatterplot depicts all earthquakes in the dataset and their related depths. A blue, horizontal, dashed line indicates earthquakes with a magnitude of 8.0 or greater.

In order to get a better idea of the distribution of these strong quakes, we created a boxplot that isolates only the strongest earthquakes.

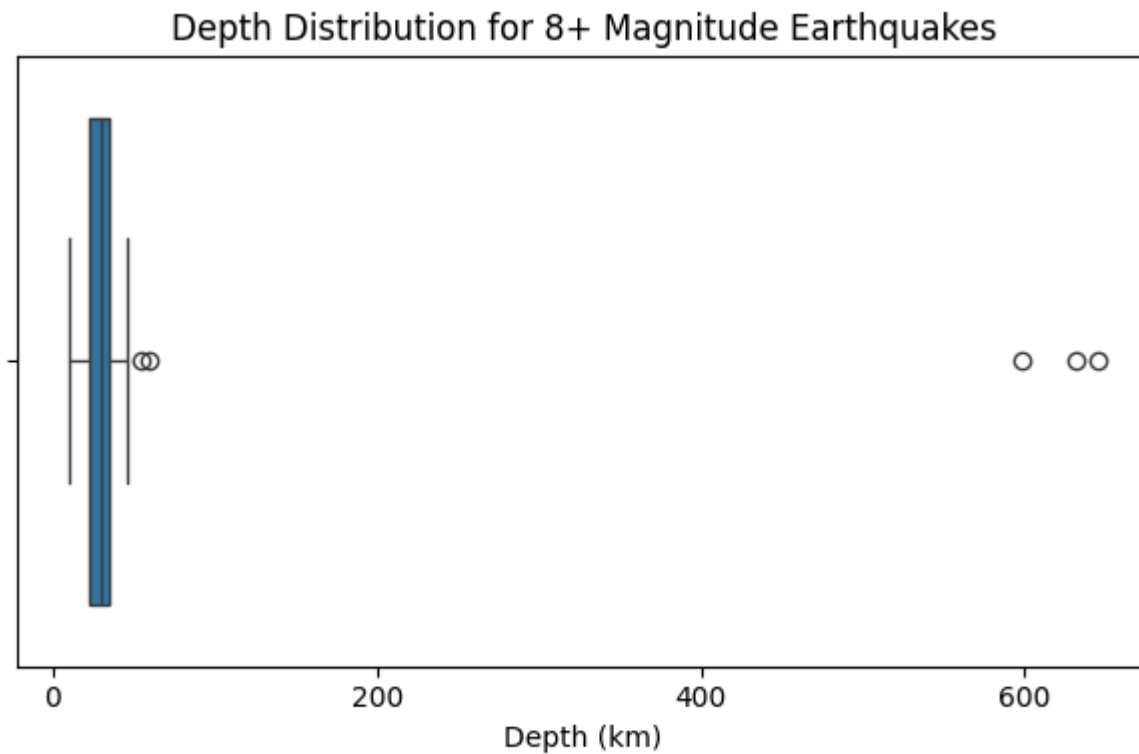
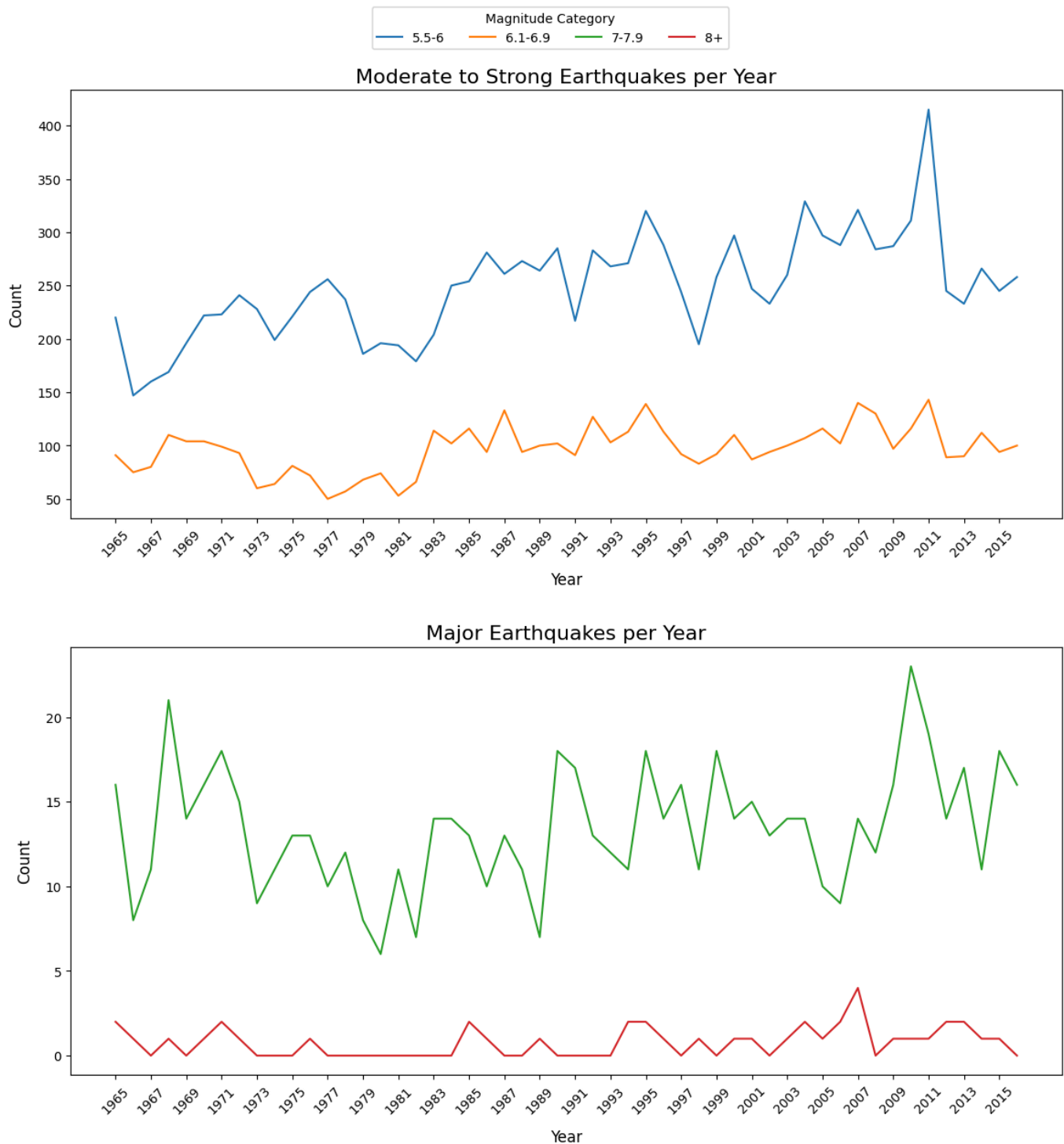


Figure 4: This boxplot shows the distribution for earthquakes and their corresponding depths. Note that this is very strongly skewed toward shallower depths, with only a few of the strongest earthquakes occurring at greater depths.

For the last portion of the EDA, we wanted to get a look at the number of earthquakes in various categories over all of the years in the dataset. This lineplot was split into two separate plots: one with earthquakes of magnitudes 5.5-6, and 6.1-6.9, and the other with earthquakes of magnitudes 7-7.9, and 8+.



The vast majority of earthquakes in this dataset are in the category with magnitudes of 5.5-6, while magnitudes of 8+ occur with far less frequency.

Machine Learning Pipeline

For this dataset we chose to use the Random Forest Classifier, which is a classifier that builds a forest of individual decision trees, allowing for better generalization on unseen data. For this ML pipeline, our target was Magnitude Category, which splits the data into categories (as seen in the EDA section) based on their magnitudes. It is worth noting that the classes have a heavy imbalance when it comes to the most extreme earthquakes. While we won't be using SMOTE to correct the class imbalance in this particular pipeline, we will discuss options for improving model metrics after establishing a baseline.

For model preprocessing, any missing numerical values were imputed for median with the Simple Imputer from the sklearn packages, and categorical columns were one-hot encoded because Random Forest requires numerical input. We used 200 estimators in the forest, a random state for reproducibility, and a balanced class weight. During the train/test split, we used a split of 80% training and 20% testing.

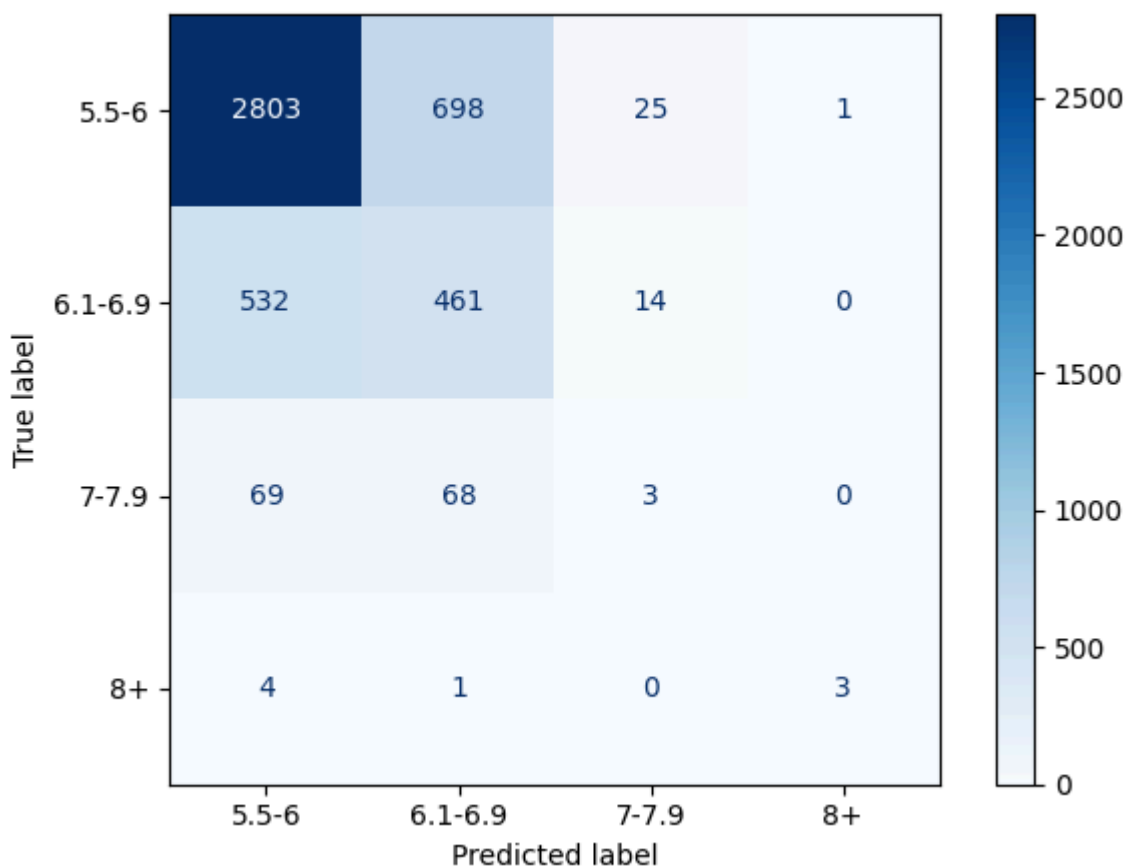
Results

The results from the model training are shown in the classification report and confusion matrix below:

Classification Report

	precision	recall	f1-score	support
5.5-6	0.822	0.795	0.808	3527.0
6.1-6.9	0.375	0.458	0.413	1007.0
7-7.9	0.071	0.021	0.033	140.0
8+	0.75	0.375	0.5	8.0
accuracy	0.698	0.698	0.698	0.698
macro avg	0.505	0.412	0.438	4682.0
weighted avg	0.704	0.698	0.7	4682.0

Confusion Matrix for Random Forest Classifier



We can see above that the model performed moderately well on the 5.5-6 magnitude category, and fairly poorly on other classes. In particular, the F1 score for the two most dangerous and destructive categories (7-7.9 and 8+) were quite low. This is worth noting because the risk for misclassification here is quite high.

Discussion

With more work, or a more complex model, in conjunction with methods for correcting class imbalance, we *may* be able to achieve a better result. That said, synthetic oversampling to correct class imbalance adds a layer of complexity to this project that is beyond a somewhat introductory machine learning class. It is also worth noting that this is a fairly common problem with natural science datasets, as there are often cases where the most rare events simply do not have enough data to bring classes back into balance.

Sources

1. Wikipedia Contributors. "Earthquake." Wikipedia, Wikimedia Foundation, 2 Mar. 2019, en.wikipedia.org/wiki/Earthquake.
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3. Contributors, Kaggle. "Significant Earthquakes, 1965-2016." [Www.kaggle.com, 2016, www.kaggle.com/datasets/usgs/earthquake-database](https://www.kaggle.com/2016).
4. Wikipedia Contributors. "List of Tectonic Plates." Wikipedia, Wikimedia Foundation, 2 June 2019, en.wikipedia.org/wiki/List_of_tectonic_plates.